

## HW4: AI-assisted data viz and extraction :)

Summary: In this 'last' HW, you'll use the magic of GenAI, to do simple ML and data viz, inside Google's Colab.

Even though it's small and quick, it will expose you new data analysis functionality 'out of the box' [no installation of anything needed!]. Going forward you can use this in your other data explorations.

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As you know, Google Colab [<https://colab.research.google.com/>] is a Jupyter Notebook environment that Google provides (along with a free TPU instance for fast computations!): using Colab you can develop and store 'notebooks' (.ipynb JSON files) that contain a mix of code cells [which can be executed] and non-code cells, eg. text headers, body, images etc. [which can't be executed], right in your Google Drive area (and share them with others, hence 'Co'lab). You can run tons of notebooks in Colab, eg. from [here](#), and play with tons of tools (add-ons) from [here](#). 

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For this hw, you are going to work with the Google Data Science Assistant (DSA) [<https://developers.googleblog.com/en/data-science-agent-in-colab-with-gemini/>] which is built right into Colab! It in turn is a Colab version of this project: <https://github.com/togethercomputer/open-data-scientist?tab=readme-ov-file>

You can alternatively use Gemini. In the following, the descriptions slightly 'favor' Gemini but you can use either Gemini or DSA (or even mix and match, eg. Q1, Q3, Q5 in Gemini, and Q2, Q4 in DSA).

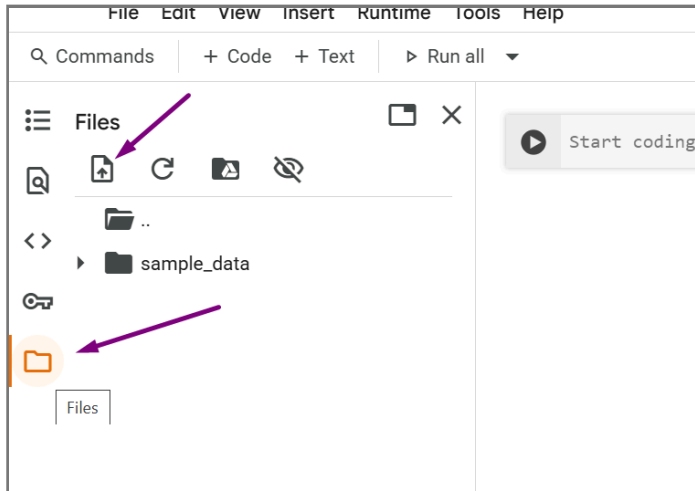
You will perform three queries on a Stack Overflow dataset, and two queries on a JSON file [which can be used as a data format, as you know].

The queries are actually, prompts to the LLM - **be sure to save them ongoing, to a text file** - because when you save your notebook to Google Drive and then reopen it, the prompts will not show up [because they are not saved with the notebook - although you could do that if you like, by creating a new 'text' cell in the notebook and pasting the prompts into it ongoing :)].

For Q1, Q2 and Q3, you'll be using the 2024 data from <https://survey.stackoverflow.co/>, ie <https://survey.stackoverflow.co/2024/>. Start by downloading the dataset at 'Download Full Data Set (CSV)', and look for 'survey\_results\_public.csv' in the stack-overflow-developer-survey-2024/

folder], which contains developer data - coding languages, age ranges, salaries, etc. Here is that same dataset [alternative dl location]: <https://www.kaggle.com/datasets/ehsansepahsalari/stack-overflow-annual-developer-survey-2024> ['Stack Overflow Annual Developer Survey2024.csv']

Bring up Colab, then do File -> New notebook in GDrive. Upload the .csv file to your notebook:



Now you can start to analyze the data in the .csv, by simply prompting Gemini via the '❖ Gemini' button at the top right of your notebook.

Q1. Ask for a histogram of developer's ages. Neat that the AI produces a pandas + matplotlib result, of the data (age ranges)! For fun, ask for other viz types, eg. a piechart

Q2. Ask DSA/Gemini to create a piechart viz, of the most popular programming language devs use. Neat - you can take a quick look at the graphic and pick out the biggest slices :)

Q3. Next, ask it to model a regression line between age and salary. Then, ask it to use the model to predict the likely salary of a 30 year old developer. Cool that DSA is able to create code build a model, build it, then use it for prediction.

Take screenshots, for Q1..Q3.

For Q4 and Q5, we will use data in JSON format, for this inexpensive but fun smartwatch: <https://www.espruino.com/Bangle.js2>

It's a microcontroller board that supports running programs in... JavaScript! Here is the spec(data) for the board, in JSON format: <https://www.espruino.com/json/BANGLEJS2.json> Take a few minutes to scroll through to see what all it offers: a lot!

Load the .json file into a **new** Colab project.

Q4. Ask Gemini/DSA to list the values for the 'pins' key. Neat - it is able to query for a value, given a key (similar to NoSQL).

Q5. Ask for values of the 'devices' key, to see what fun devices the board comes with.

Get Q4, Q5 screenshots.

That's it, you're done :) The intent here is to have you play with DSA, to see how easy and non-technical data analysis can be. Note that you can 'keep' your notebooks for future study/use (they are auto-saved anyway), and, modify the generated code or if you want, or make the prompt be

more detailed or specific [which vibe-coders and non-DS people can't do!].

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Please post your qns (and their answers!) on the 'hw4' Piazza page.

**What to submit** (as a single .zip):

- screenshots all the Q1..Q5 questions
- a transcript.txt text file of the entire (Q1..Q5) chat session, ie. what you typed, what Gemini output. It's ok if the transcript is 'messy' [contains trial prompts you put in, things you entered that were not asked for in Q1..Q5, etc].

**HAVE FUN!**

